FYUGP B.Sc. PHYSICS HONOURS

ELECTRODYNAMICS II

Semester V · Core in Major

MODULE 1: ELECTRIC AND MAGNETIC FIELDS IN MATTER

Model questions on topics the past papers miss

Section A: 1 mark each

Polarization

1 Write the relation between the induced dipole moment $\mathbf{p}$ of an atom and the field $\mathbf{E}$. Give the polarizability of a model atom of radius a. **(1)**

2 Write the bound volume charge density ρ_b in terms of the polarization $\mathbf{P}$. **(1)**

3 Write the field inside a uniformly polarized sphere of radius R. **(1)**

Magnetization

4 Write the torque on a magnetic dipole $\mathbf{m}$ in a uniform field $\mathbf{B}$. **(1)**

5 Write the bound surface current $\mathbf{K}_b$ in terms of the magnetization $\mathbf{M}$ and the normal $\hat{\mathbf{n}}$. **(1)**

6 Write $\mathbf{B}$ in terms of $\mathbf{H}$ for a linear magnetic medium of susceptibility χ_m. **(1)**

Section B: 2 marks each, answer all

Polarization and bound charge

7 Explain physically why a string of head-to-tail dipoles leaves net charge at its two ends. **(2)**

8 State the boundary condition on $D^{\perp}$ and on $\mathbf{D}^{\parallel}$ at an interface, in terms of free charge and polarization. **(2)**

Magnetization and bound currents

9 Explain physically why a uniformly magnetized slab carries a surface current, although each atom carries only a tiny loop current. **(2)**

10 Show that $\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}$ and explain why $\mathbf{H}$ cannot be found from the free current alone in a bar magnet. **(2)**

11 Explain why a magnetic dipole in a uniform field feels a torque but no net force, and why this changes in a non-uniform field. **(2)**

Section C: 4 marks each, answer any two

Polarization

12 A point charge $Q = 1\,\mu$C is fixed at the origin. A dipole $p = 1$ nC m is free to ride on a circle of radius $R = 0.1$ m about the charge, with its axis tangent to the circle. Using $\mathbf{F} = (\mathbf{p} \cdot \nabla)\mathbf{E}$, show that the force on the dipole is $\mathbf{F} = \dfrac{Q\,\mathbf{p}}{4\pi\varepsilon_0 R^3}$ and find its magnitude. **(4)**

13 A sphere of radius $R = 0.1$ m carries polarization $\mathbf{P} = k\mathbf{r}$, with $k = 2 \times 10^{-6}$ C/m^2. Find the bound volume charge ρ_b, the bound surface charge σ_b, and the total bound charge. **(4)**

14 A solid sphere of radius 2 cm is uniformly polarized with $P = 1.0 \times 10^{-6}$ C/m^2. Find the field inside the sphere and the dipole moment of the sphere. Take $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m. **(4)**

15 A dielectric sphere of relative permittivity $\varepsilon_r = 3$ is placed in a uniform field $E_0 = 1000$ V/m. Find the field inside the sphere and the polarization inside it. **(4)**

Magnetization

16 A uniformly magnetized sphere of radius 5 cm has $M = 1.0 \times 10^4$ A/m. Find the magnetic field inside the sphere and its magnetic dipole moment. Take $\mu_0 = 4\pi \times 10^{-7}$ H/m. **(4)**

17 A long solenoid with $n = 1000$ turns/m carries a current $I = 2$ A and is filled with a linear material of $\chi_m = 2 \times 10^{-5}$. Find $\mathbf{H}$, $\mathbf{B}$ inside, and the bound surface current density on the surface of the material. **(4)**

Section D: 10 marks each, answer any one

Polarization

18 Starting from the bound charges $\rho_b = -\nabla \cdot \mathbf{P}$ and $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ of a uniformly polarized sphere, derive the field inside the sphere and the potential outside it. Show that the outside field is that of a dipole $\mathbf{p} = \frac{4}{3}\pi R^3 \mathbf{P}$. **(10)**

19 A conducting sphere of radius a carries a charge Q and is surrounded by linear dielectric of permittivity ε out to radius b. Find $\mathbf{D}$, $\mathbf{E}$ and the potential at the centre of the sphere, relative to infinity. **(10)**

Magnetization

20 Explain diamagnetism using the orbital motion of an electron in a magnetic field. Derive the change in the orbital dipole moment $\Delta\mathbf{m} = -\dfrac{e^2 R^2}{4m_e}\mathbf{B}$, and explain why it is opposite to $\mathbf{B}$ and weaker than paramagnetism. **(10)**

MODULE 1: MODEL ANSWERS

For checking your work. Each answer shows the key step and the result.

1 $\mathbf{p} = \alpha\mathbf{E}$; $\alpha = 4\pi\varepsilon_0 a^3$. **2** $\rho_b = -\nabla\cdot\mathbf{P}$. **3** $\mathbf{E} = -\mathbf{P}/3\varepsilon_0$. **4** $\boldsymbol{\tau} = \mathbf{m}\times\mathbf{B}$. **5** $\mathbf{K}_b = \mathbf{M}\times\hat{\mathbf{n}}$. **6** $\mathbf{B} = \mu_0(1+\chi_m)\mathbf{H}$.

7 Positive charges at one end and negative at the other: each internal head cancels the next tail, but the two ends are left uncancelled. **8** $D^{\perp}_{above} - D^{\perp}_{below} = \sigma_f$; $\mathbf{D}^{\parallel}_{above} - \mathbf{D}^{\parallel}_{below} = \mathbf{P}^{\parallel}_{above} - \mathbf{P}^{\parallel}_{below}$. **9** Each atom gives a loop current, but at the edge there is no neighbour to cancel it, so the loops add up to one surface current $K_b = M$. **10** $\nabla\times\mathbf{H} = \mathbf{J}_f$ but $\nabla\cdot\mathbf{H} = -\nabla\cdot\mathbf{M} \neq 0$ at the ends of a bar magnet; H is found from symmetry and free current only where $\nabla\cdot\mathbf{M} = 0$. **11** A uniform field gives $\mathbf{F} = 0$ but $\boldsymbol{\tau} = \mathbf{m}\times\mathbf{B}$; in a non-uniform field $\mathbf{F} = \nabla(\mathbf{m}\cdot\mathbf{B}) \neq 0$.

12 Tangent dipole, $\mathbf{p}\cdot\nabla$ along the tangent gives $F = \dfrac{Qp}{4\pi\varepsilon_0 R^3} = 8.99\times10^{-3}$ N (about 9.0 mN), tangential. **13** $\rho_b = -3k = -6\times10^{-6}$ C/m^3; $\sigma_b = kR = 2\times10^{-7}$ C/m^2; total $= -4\pi kR^3 + 4\pi kR^3 = 0$. **14** $E = -P/3\varepsilon_0 = -3.77\times10^{4}$ V/m (magnitude 3.77×10^{4} V/m); $p = \frac{4}{3}\pi R^3 P = 3.35\times10^{-11}$ C m. **15** $E = 3E_0/(\varepsilon_r+2) = 600$ V/m; $\mathbf{P} = \varepsilon_0\chi_e\mathbf{E}$ with $\chi_e = 2$ gives $P = 1.06\times10^{-8}$ C/m^2. **16** $B = \frac{2}{3}\mu_0 M = 8.38\times10^{-3}$ T; $m = \frac{4}{3}\pi R^3 M = 5.24$ A m^2. **17** $H = nI = 2000$ A/m; $B = \mu_0(1+\chi_m)H = 2.51\times10^{-3}$ T; $K_b = \chi_m H = 0.04$ A/m.

18 Inside, ρ_b is uniform and $\sigma_b = P\cos\theta$ on the surface; the potential is $V_{in} = \frac{P}{3\varepsilon_0} r\cos\theta$ and $\mathbf{E}_{in} = -\mathbf{P}/3\varepsilon_0$. Outside, $V = \frac{1}{4\pi\varepsilon_0}\frac{\mathbf{p}\cdot\hat{\mathbf{r}}}{r^2}$ with $\mathbf{p} = \frac{4}{3}\pi R^3\mathbf{P}$. **19** Gauss gives $\mathbf{D} = \dfrac{Q}{4\pi r^2}\hat{\mathbf{r}}$ everywhere; $\mathbf{E} = \dfrac{Q}{4\pi\varepsilon r^2}\hat{\mathbf{r}}$ for $a < r < b$. Potential at the centre $V = \dfrac{Q}{4\pi}\left(\dfrac{1}{\varepsilon_0 b} + \dfrac{1}{\varepsilon a} - \dfrac{1}{\varepsilon b}\right)$. **20** Larmor: an electron orbit speeds up or slows down in B, so $\Delta v = eRB/2m_e$ and $\Delta\mathbf{m} = -\frac{1}{2}e\,\Delta v\,R\,\hat{z}$, opposite to B for both senses of orbit. Diamagnetism acts on all atoms, but paramagnetic torques on unpaired spins are much larger, so it is weaker.

How the model paper was made

The Module 1 papers in `source/pyq` repeat a small set of textbook items: the polarized-object potential (Eq. 4.8–4.13), the dielectric constant and Clausius–Mossotti (Eq. 4.72), Gauss's law with a dielectric (Eq. 4.20–4.23), the Rowland ring and toroid gap (Ex. 6.2 and Eq. 6.18–6.20), and the ferromagnet domains and hysteresis (Sec. 6.4.2). They ask definitions, one-line relations and a long derivation. The topics below appear in none of the 39 Module 1 questions.

Missed topic	Griffiths 5th (from the excerpt)	Model questions
Induced dipole and atomic polarizability	Sec. 4.1.2; Eq. 4.1, 4.2 (p.167–168)	1
Bound volume charge, $\rho_b = -\nabla\cdot\mathbf{P}$	Sec. 4.2.1; Eq. 4.12 (p.173)	2, 13
Bound charge: physical picture	Sec. 4.2.2; Fig. 4.11 (p.175–177)	7
Uniformly polarized sphere: field and dipole	Ex. 4.2; Eq. 4.14–4.16 (p.174)	3, 14, 18
Force on a dipole in a non-uniform field	Sec. 4.1.3; Eq. 4.5 (p.170), Prob. 4.31 (p.205)	12
D boundary conditions	Sec. 4.3.3; Eq. 4.26–4.27 (p.184)	8
Metal sphere in dielectric shell	Ex. 4.5; Eq. 4.35 (p.188–189)	19
Dielectric sphere in uniform field, $E_{in} = 3E_0/(\varepsilon_r+2)$	Ex. 4.7; Eq. 4.49 (p.193–194)	15
Atomic polarizability from ε_r (Clausius–Mossotti)	Prob. 4.42; Eq. 4.72 (p.207–208)	—
Magnetic dipole torque and force	Sec. 6.1.2; Eq. 6.1, 6.3 (p.269–271)	4, 11
Diamagnetism from electron orbits	Sec. 6.1.3; Eq. 6.4–6.8 (p.273–274)	20
Uniformly magnetized sphere: $B = \frac{2}{3}\mu_0 M$	Ex. 6.1; Eq. 6.16 (p.277–278)	16
Bound surface current $\mathbf{K}_b = \mathbf{M}\times\hat{\mathbf{n}}$	Sec. 6.2.2; Eq. 6.14 (p.279–280)	5, 9
$\nabla\cdot\mathbf{H} = -\nabla\cdot\mathbf{M}$	Sec. 6.3.2; Eq. 6.23 (p.285)	10
Linear magnetic medium: $\mathbf{B} = \mu_0(1+\chi_m)\mathbf{H}$, solenoid	Ex. 6.3; Eq. 6.30–6.32 (p.287–288)	6, 17

These are model questions written for this kit, not past papers. Equation numbers and pages come from the Module 1 excerpt in `source/books`. Problem 4.42 (Clausius–Mossotti) is in the excerpt but not used in the paper. The B and H boundary conditions (Sec. 6.3.3) are left out, per decision D3.